The DNA of Numbers

From linear continuity to spiral progression

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Foreword

This document is a continuation of my earlier work. Where the previous exploration focused on the linear equation N = pA + qB, on the clock, and on the connection between the spiral and the digital root (with units of time as a concrete example); this text goes one layer deeper.

The question is now not only where a number stands on the spiral, but how it relates to the two steps that produce it. The linear equation N = pA + qB thereby becomes the foundation of the spirals: the same digital root that determined the height there determines the smallest coordinate here.

The theory that follows is verifiable, reproducible, and based entirely on elementary number theory. Every example can be checked by hand. The poetry is not in the numbers themselves, but in the structure they reveal.

Introduction

Numbers are not silent objects. They carry a structure within them that only becomes visible when you question them in the right way. This document poses one simple question: how often can a number N be written as a weighted sum of two steps p and q? It follows what that question yields.

The answer turns out to be surprisingly rich. The smallest coordinate of every solution is completely determined by the digital root of N. The number of solutions follows a closed formula. And the numbers that appear anomalous turn out to be the key to a deeper structure: the family 1675–1683, in which all nine digital roots occur exactly once and every number reaches its full representation count, except one.

That single exception, the number 1682, is not a flaw in the system. It is the fingerprint of the residue class. And that fingerprint leads us back to the spirals: the same digital root that determines the smallest coordinate here determined the height of the spiral there. Two theories, one function.

This document has three layers:

1. The linear equation N = pA + qB with p ≡ 1 (mod q): the algebraic basis.
2. The three classes of digital roots, determined by the divisibility of q by 3 and 9: the structural layer.
3. The family 1675–1683 and the bridge to the spirals: the visual layer.

1. The core formula

The entire system rests on one linear Diophantine equation:

$$N = pA + qB$$

with p, q positive integers, gcd(p,q)=1, $p > q \geq 1$, and the main condition:

$$p \equiv 1 \pmod{q}$$

The variables A, $B \in \mathbb{Z}_{\geq 0}$ are the coordinates of N. A representation of N is a pair (A,B) that satisfies the equation.

In the special case (p,q) = (19,9) this becomes:

$$N = 19A + 9B$$

Because $19 \equiv 1 \pmod{9}$, it holds for every representation that:

$$N \equiv pA \equiv A \pmod{q}$$

Because of the condition $p \equiv 1 \pmod{q}$, every solution satisfies $A \equiv N \pmod{q}$. The smallest non-negative coordinate A_0 is therefore the least non-negative residue of N modulo q:

$$A_0 = N \bmod q$$

When $N \equiv 0 \pmod{q}$ we have $A_0 = 0$. For q = 9 the digital root is the related quantity $1 + ((N-1) \bmod 9)$, which maps multiples of 9 to 9 rather than 0. The successive representations then arise from the step vector:

$$A \to A + q \ , \ B \to B - p$$

and the number of representations is given by the closed formula:

$$R(p,q;N) = \left\lfloor \left(\lfloor N/p \rfloor - A_0\right) / q \right\rfloor + 1$$

The family bound (the smallest number with at least k representations) is:

$$G(k) = pq(k-1) + (p-1)(q-1)$$

For (19,9): G(k) = 171k − 27. Thus G(10) = 1683.

What this means. Every number has a unique smallest coordinate A_0 (the least non-negative residue of N modulo q). For q = 9 this coincides with the digital root except when N is a multiple of 9, in which case $A_0 = 0$ while the digital root is 9. The representations lie on an arithmetic progression with step q. And the threshold at which a number reaches its full number of representations lies at a fixed distance pq. The system is fully calibrated.

2. The three classes

The digital root of p = 1 + qm behaves in exactly three ways, determined by the divisibility of q by 3 and 9:

Class	Condition	Cycle of d	Examples
A	q ≢ 0 (mod 3)	full cycle of 9 (permutation of 1–9)	q = 2, 4, 5, 7, 8, 10, 365
B	q ≡ 0 (mod 3), q ≢ 0 (mod 9)	short cycle of 3, trapped in {1,4,7}	q = 3, 6, 12, 15, 30, 60, 120, 240
C	q ≡ 0 (mod 9)	constant d = 1 (flat spiral)	q = 9, 18, 27, 36, 360, 3600, 86400

The divisibility of q by 3 and 9 determines the category.

Class A is the richest class: the digital root runs through all nine levels. The week (q=7) is the best-known example: each day brings a new digital root, and after seven days the cycle begins again, but one level higher. Class B is the class of time: 60 minutes, 60 seconds, 24 hours, 12 months; they remain trapped in {1,4,7}. Class C is the class of the flat spiral: 360°, 3600 seconds, 86400 seconds; the digital root stays constantly 1. There is no rise.

3. The ten levels per modulus (q = 9 to q = 2)

Below are the ten levels for each modulus q, from 9 down to 2. For each modulus q we define p = 1 + qm, and compute the digital root d of p. The table shows for the first ten levels m = 0, 1, …, 9: the position p, the residue p mod 9, and the digital root d = dr(p).

The category is determined by q:

· q ≡ 0 (mod 9) → Class C (constant)

· q ≡ 0 (mod 3), q ≢ 0 (mod 9) → Class B (cycle of 3)

· q ≢ 0 (mod 3) → Class A (cycle of 9)

q = 9 — Class C (constant)

p = 1 + 9m, so p ≡ 1 (mod 9) for all m. The digital root is always 1.

m	p = 1 + 9m	p mod 9	d = dr(p)
0	1	1	1
1	10	1	1
2	19	1	1
3	28	1	1
4	37	1	1
5	46	1	1
6	55	1	1
7	64	1	1
8	73	1	1
9	82	1	1

Observation: the spiral is flat. There is no rise. Every step of 9 brings you back to the same digital root. This is the class of 360°, 3600 seconds, 86400 seconds.

q = 8 — Class A (full cycle of 9)

p = 1 + 8m. Because gcd(8,9) = 1, p mod 9 runs through all nine residues.

m	p = 1 + 8m	p mod 9	d = dr(p)
0	1	1	1
1	9	0	9
2	17	8	8
3	25	7	7
4	33	6	6
5	41	5	5
6	49	4	4
7	57	3	3
8	65	2	2
9	73	1	1

Observation: after 9 steps the digital root is back at 1, but the position has risen from 1 to 73. The spiral has completed one full turn and is one level higher.

q = 7 — Class A (full cycle of 9)

p = 1 + 7m. Because gcd(7,9) = 1, p mod 9 runs through all nine residues.

m	p = 1 + 7m	p mod 9	d = dr(p)
0	1	1	1
1	8	8	8
2	15	6	6
3	22	4	4
4	29	2	2
5	36	0	9
6	43	7	7
7	50	5	5
8	57	3	3
9	64	1	1

Observation: the week is the best-known example of Class A. Each day brings a new digital root, and after seven days the cycle begins again, but one level higher. The order is 1, 8, 6, 4, 2, 9, 7, 5, 3, 1. No digital root is skipped.

q = 6 — Class B (cycle of 3)

p = 1 + 6m. Because gcd(6,9) = 3, p mod 9 remains trapped in {1, 4, 7}.

m	p = 1 + 6m	p mod 9	d = dr(p)
0	1	1	1
1	7	7	7
2	13	4	4
3	19	1	1
4	25	7	7
5	31	4	4
6	37	1	1
7	43	7	7
8	49	4	4
9	55	1	1

Observation: the cycle is 1, 7, 4, 1, 7, 4, … The digital root is trapped in three values. This is the class of 60 minutes, 60 seconds, 24 hours, 12 months. Time remains stuck in {1, 4, 7}.

q = 5 — Class A (full cycle of 9)

p = 1 + 5m. Because gcd(5,9) = 1, p mod 9 runs through all nine residues.

m	p = 1 + 5m	p mod 9	d = dr(p)
0	1	1	1
1	6	6	6
2	11	2	2
3	16	7	7
4	21	3	3
5	26	8	8
6	31	4	4
7	36	0	9
8	41	5	5
9	46	1	1

Observation: the order is 1, 6, 2, 7, 3, 8, 4, 9, 5, 1. All nine digital roots occur. The spiral rises.

q = 4 — Class A (full cycle of 9)

p = 1 + 4m. Because gcd(4,9) = 1, p mod 9 runs through all nine residues.

m	p = 1 + 4m	p mod 9	d = dr(p)
0	1	1	1
1	5	5	5
2	9	0	9
3	13	4	4
4	17	8	8
5	21	3	3
6	25	7	7
7	29	2	2
8	33	6	6
9	37	1	1

Observation: the order is 1, 5, 9, 4, 8, 3, 7, 2, 6, 1. All nine digital roots occur. The spiral rises.

q = 3 — Class B (cycle of 3)

p = 1 + 3m. Because gcd(3,9) = 3, p mod 9 remains trapped in {1, 4, 7}.

m	p = 1 + 3m	p mod 9	d = dr(p)
0	1	1	1
1	4	4	4
2	7	7	7
3	10	1	1
4	13	4	4
5	16	7	7
6	19	1	1
7	22	4	4
8	25	7	7
9	28	1	1

Observation: the cycle is 1, 4, 7, 1, 4, 7, … The digital root is trapped in three values. This is the class of the trimester, the quarter, and all moduli divisible by 3 but not by 9.

q = 2 — Class A (full cycle of 9)

p = 1 + 2m. Because gcd(2,9) = 1, p mod 9 runs through all nine residues.

m	p = 1 + 2m	p mod 9	d = dr(p)
0	1	1	1
1	3	3	3
2	5	5	5
3	7	7	7
4	9	0	9
5	11	2	2
6	13	4	4
7	15	6	6
8	17	8	8
9	19	1	1

Observation: the order is 1, 3, 5, 7, 9, 2, 4, 6, 8, 1. All nine digital roots occur. The spiral rises. This is the class of the day/night cycle (q=2), and all moduli not divisible by 3.

4. The family 1675–1683 — the closing example

The nine consecutive numbers 1675–1683 form the first contiguous block in which each of the nine digital roots 1–9 occurs exactly once, and in which every number has reached its full representation count, except one: 1682.

The table below uses the equation 19A + 9B = N. The column “symmetry point” contains a pair (A,B) that is both a valid representation and satisfies dr(A) = dr(B) = dr(N).

N	reduction chain	dr(N)	A_0	R(N)	symmetry point (A,B)
1675	1675 → 19 → 10 → 1	1	1	10	(28,127)
1676	1676 → 20 → 2	2	2	10	(2,182) and (83,11)
1677	1677 → 21 → 3	3	3	10	(57,66)
1678	1678 → 22 → 4	4	4	10	(31,121)
1679	1679 → 23 → 5	5	5	10	(5,176) and (86,5)
1680	1680 → 15 → 6	6	6	10	(60,60), A=B
1681	1681 → 16 → 7	7	7	10	(34,115)
1682	1682 → 17 → 8	8	8	9	(8,170)
1683	1683 → 18 → 9	9	0	10	(63,54)

Verification of the symmetry points:

- 19·28 + 9·127 = 532 + 1143 = 1675
- 19·60 + 9·60 = 1140 + 540 = 1680 (the point where A=B)
- 19·8 + 9·170 = 152 + 1530 = 1682

Important for the reader: the numbers in the symmetry points are not ordinary sums. The pair (28,127) does not add up to 1675 via 28+127, but via 19·28 + 9·127. The coordinates are weighted by p and q.

5. Why 1682 has one representation fewer

The family bound for k=10 is:

$$G(10) = 171 \cdot 10 - 27 = 1683$$

From 1683 onward every number has at least 10 representations. The number 1682 lies one below this bound and therefore has R = 9. That is a structural consequence of the residue class.

The reasoning:

1. The representations of N are A_0, A_0+9, A_0+18, … up to ⌊N/19⌋.
2. The number of steps is ⌊(⌊N/19⌋ − A_0)/9⌋ + 1.
3. For N = 1682: ⌊1682/19⌋ = 88, A_0 = 8, so R = ⌊(88−8)/9⌋ + 1 = ⌊80/9⌋ + 1 = 8+1 = 9.
4. For N = 1683: ⌊1683/19⌋ = 88. Because 1683 ≡ 0 (mod 9), the smallest coordinate is A_0 = 0. The closed formula then gives R = ⌊(88 − 0)/9⌋ + 1 = ⌊88/9⌋ + 1 = 9 + 1 = 10. The pair (A, B) = (0, 187) is a valid representation, and all ten representations are: A = 0, 9, 18, 27, 36, 45, 54, 63, 72, 81.

Numbers with N ≡ 8 (mod 9) reach the family bound one step later than their neighbours. In 1682 this manifests as R = 9 instead of 10: the digital root 8 lies just before the threshold G(10) = 1683. The residue class determines the distance to the bound.

The 8 is not a signal being sent again. It is simply a position on the number line that lies one step before the threshold. The structure resides in the threshold, not in the digit 8.

6. The bridge to the spirals

The digital root is the same function in both worlds:

$$d = 1 + ((p-1) \bmod 9) \leftrightarrow A_0 = 1 + ((N-1) \bmod 9)$$

In the spiral theory, d is the height of the spiral (the level). In the representation system, A_0 is the smallest coordinate. It is the same quantity, viewed twice:

· Spiral: p = 1 + qm, and d determines which step you stand on.

· Representation: N = pA + qB, and A_0 = dr_q(N) determines the smallest A.

The family 1675–1683 shows this in a single block: nine consecutive numbers, each on a different step of the spiral (because each has a different digital root).